

HYBRID IMAGE ENHANCEMENT WITH DEEP LEARNING FOR DIABETIC RETINOPATHY DETECTION

ABSTRACT

If not detected early, diabetic retinopathy (DR), a dangerous retinal disorder that can arise from long-term diabetes mellitus, is one of the main causes of vision loss. For DR to be successfully treated and to avoid of blindness. Such a system might offer useful information to aid in diagnosis given the growing interest in contemporary methods for retinal fundus image analysis; however, low contrast and unclear clarity could reduce diagnostic reliability. Most earlier research has concentrated on image enhancement techniques to improve detection performance, which highlights the need for a more flexible and reliable diagnostic decision-making approach. This study proposes a novel, efficient framework for diagnosing pathological signs of DR by combining deep learning-based classification, adaptive preprocessing, and precise retinal region segmentation. First, contrast-limited adaptive histogram equalization (CLAHE), clarification normalization, and green channel extraction are used to improve the quality of retinal images. The EfficientNet-B2 model then extracts and categorizes informative features. The suggested approach outperforms a number of earlier enhancement algorithms with a classification accuracy of 85.8% when tested on common benchmark datasets like APTOS. The suggested method can be used to detect diabetic retinopathy in Internet of Medical Things (IoMT)-centric diagnostic systems and smart healthcare settings.

Keywords: fundus images, diabetic retinopathy (DR), deep learning, image enhancement

I. INTRODUCTION

One of the main causes of vision impairment worldwide is diabetic retinopathy (DR), a chronic ocular condition brought on by ongoing diabetes mellitus. Over time, the retina's structure and function deteriorate due to the gradual damage caused by prolonged elevation of blood glucose levels to the retinal blood vessels. The disease frequently advances without obvious symptoms in its early stages, delaying clinical diagnosis and raising the risk of severe visual impairment or irreversible blindness if prompt treatment is not given. Early detection and routine monitoring of diabetic retinopathy have become vitally important due to the ongoing global increase in the prevalence of diabetes. A crucial diagnostic tool, retinal imaging allows for the fine-grained visualization of retinal characteristics like vascular alterations and lesion formations linked to the progression of disease. However, traditional diagnostic methods primarily rely on manual examination by specialists in ophthalmology, which makes the process laborious and prone to arbitrary interpretation, especially when used in mass screening programs.

Computerized analysis of images has witnessed significant prominence as an effective solution to these drawbacks, which can facilitate the analysis of retinal images in an objective, rapid, and constant manner. Some limitations in image analysis, including uneven illumination, image noise, device types, etc., make it difficult to accurately grade diabetic retinopathy. As it is critical to accurately classify images across different stages of DR, there is a strong need for efficient systems that can support the analysis of DR.

SYMPTOMS

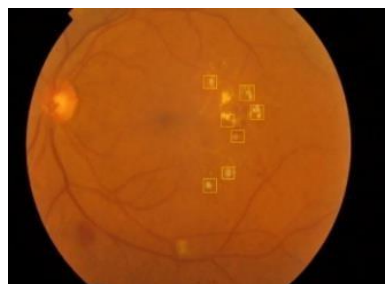
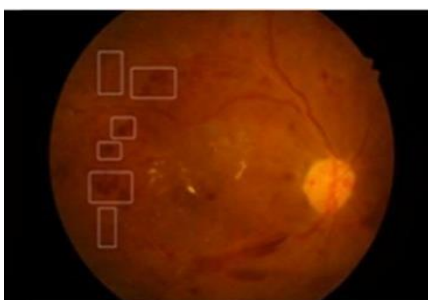


Fig 2 HEMORRHAGES

Fig 1 MICROANEURYSMS

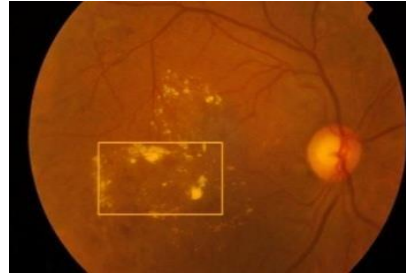
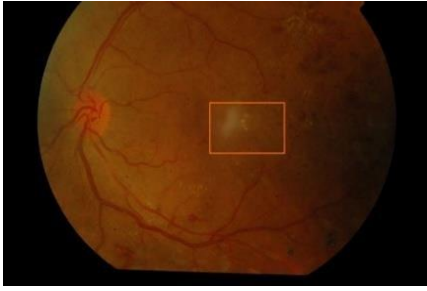


Fig 3 EXUDATES

Fig 4 COTTON WOOL

"Diabetic Retinopathy (DR)" develops in the form of some specific "retinal lesions" and such changes in the retina are important for the diagnosis of DR and the determination of the severity of the disease. The main "pathological changes" that occur during DR are the development of microaneurysms (Fig. 1), "retinal hemorrhages with vascular changes" (Fig. 2), "exudates" (Fig. 3), and "cotton wool spots" (Fig. 4). The microaneurysms occur in the early stages of diabetic retinopathy, and they appear in the retina as small, discrete, and bright red-colored dots. This occurs due to the dilation of the "retinal capillaries." "Hemorrhages occur due to the rupture of the weakened capillaries." This condition causes dark red dot-shaped or bloated structures to appear in the retina. Exudates appear in the retina in the form of bright "yellow or white dots." Cotton wool spots appear as pale, whitish spots due to localized "retinal ischemia" resulting from the occlusion of small "arterioles". These types, number, and distribution of lesions serve as the basis for severity grading of diabetic retinopathy, which warrants the precise discovery of these lesions to aid clinicians as well as to develop computer-based systems for screening the disease. (a1, a2, a3) The original images of the fundus of the retinal surfaces of the eyes(b1,b2,b3) Corresponding images after applying the enhancement technique like green channel, Illumination, showing clear retinal features.

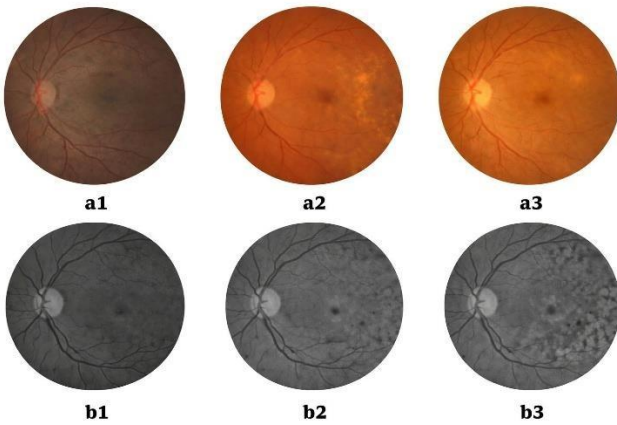


Fig 5: Sample Images

II. LITERATURE REVIEW

In the year 2022, the research targeted the enhancement of the analysis of the human retinal structures that are integral for diabetic retinopathy disease diagnosis, such as the retinal blood vessels. The researchers used a hybrid clustering-based approach using evolutionary optimization in combination with unsupervised learning methods as a potential tool for analyzing the segmentation of the retinal blood vessel in the fundus image acquired from patients suffering from diabetic retinopathy [1]. According to the researchers, the segmentation of blood vessels in the eye retina is significant for fully understanding the disease advancement, considering the fact that the abnormal functioning of the blood vessels is the initial symptom of diabetic retinopathy. The complexity in the blood vessel structures also acts as a drawback for the accuracy of the DR disease classification algorithms. In the year 2023, the focus was on the classification of diabetic retinopathy using deep learning techniques for the feature extraction processing from the acquired fundus image. The researchers used the hybrid CNN framework extract selected the features of retinal images utilizing the combination of various DNN models for the classification purpose [2]. The research proved that the integration of various features of their models increases the efficiency of the classification process. Although the authors were successful in their objective, they highlighted that the detection of DR at the initial stage is challenging, as lesions

of disease may not be significant, and there may be significant similarities in the diseases. In the year 2024, various research studies were carried out to develop systems to resolve the issues related to the detection of severity level in DR and lesions. Thus, the study suggested the development of a hybrid model that integrates efficient models of feature extraction as well as encoder-decoder models to carry out the segmentation of DR of multiple severity levels [3]. According to the literature, it was highlighted that diabetes, in its initial stages, may not cause significant visual symptoms, hence making it challenging to develop efficient models to category severity level. The study on lesion-based detection of DR highlighted the limitations of research work carried out earlier, which identified the common lesions in the context of the disease, and the diseases identified were related to high-level pathological features [4]. This study placed greater emphasis on the importance of complete retinal assessment by covering all aspects of normal and severe abnormalities for enhanced clinical relevance. In 2025, research trends are shifting to progress towards quality, transparency, and robustness of DR diagnostic systems. In a study published during this year, researchers placed great importance on retinal image enhancement. This process involves retinal image enhancement due to poor contrast and quality of fundus images that impact diagnosis severely [5]. In this context, hybrid Deep Learning (DL) models were used to improve retinal images to detect nonproliferative diabetic retinopathy at its early stage. Similarly, in another study published this year, a traditional machine learning-based classifier was used to reduce human involvement [6]. Further investigations based on these paradigms for class imbalance and the difficulty in the visibility of lesions were conducted in 2025. A conditional generative framework was introduced to efficiently handle the representation of underrepresented disease classes for prediction models related to severity level [7]. Moreover, investigations based on the edge-based deployment of the diagnostic system underscored the importance of transparency and interpretability in machine-based diagnostic tools [8]. These investigations emphasized the problem of a lack of interpretability in the system, oblivious to the level of accuracy obtained with the proposed solution. More recent investigations proposed efficient hybrid models with properties to preserve spatial relationships present in the retinal image [9]. These investigations showed improved efficacy in detecting the disease's early stages at a minimal computational cost. Overall, research efforts to create dependable and effective automated systems for diabetic retinopathy detection are evident in the literature published between 2022 and 2025. Even with these developments, there are still a number of difficulties, especially in recognizing early-stage pathological features consistently and accurately.

BLOCK DIAGRAM

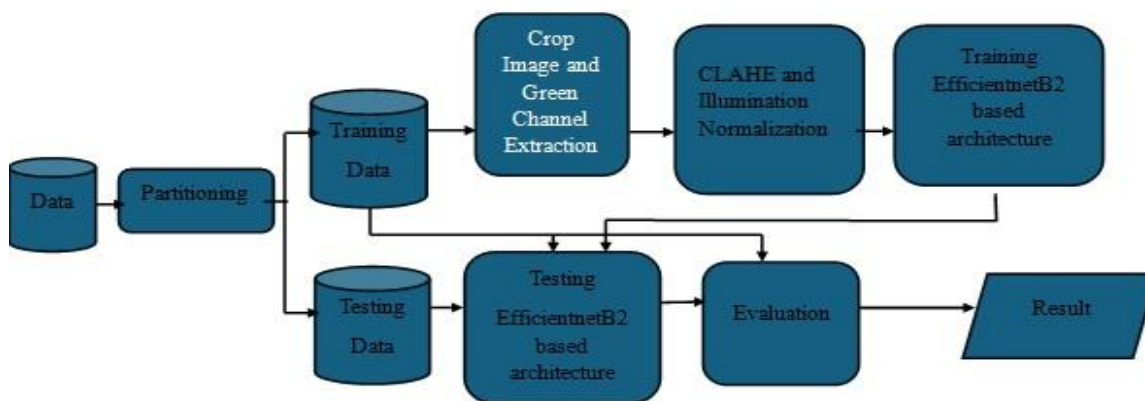


Fig 6 : Proposed Working model

The entire processing pipeline of the suggested Diabetic Retinopathy (DR) classification framework, created with the EfficientNet-B2 architecture, is shown in Figure 6. In order to facilitate accurate performance evaluation, the workflow starts with the input dataset, which is methodically separated into training and testing subsets using a data partitioning module. After that, the training images go through a number of preprocessing steps. Green channel extraction is used to highlight important retinal structures after retinal images have first been cropped to maintain the area of clinical significance. To enhance visual clarity and lessen the effects of uneven lighting, contrast enhancement is then carried out using Contrast-Limited Adaptive Histogram Equalization (CLAHE) in conjunction with illumination normalization. The deep learning architecture based on EfficientNet-B2 is then trained using the improved images for feature learning and classification. In parallel, the testing data is passed through the same EfficientNet-B2 model for prediction without affecting the trained parameters. The outputs obtained from the testing phase are forwarded to the evaluation module, where performance metrics are computed to measure the efficiency of the model. Finally, the evaluated outcomes are presented as the result, demonstrating the classification performance of the proposed system.

Algorithm 1 Crop Image and Green Channel Extraction

Input:

(1) image (2)

Threshold

Output:

GreenImage

```
1: start algorithm
2: if image.ndim == 3 then
3:   grayImage = convertColor(image, RGB2GRAY)
4:   mask = grayImage > Threshold
5:   croppedImage = image[mask.any(1), mask.any(0)]
6: else
7:   croppedImage = image
8: end if
9: greenChannel = croppedImage[:, :, 1]
10: GreenImage = greenChannel
11: end algorithm
```

A.CROPIIMAGE AND GREEN CHANNEL EXTRACTION

The proposed method takes an image of the retina fundus as input. RGB color image given as input it will be converted to the grayscale model. A threshold method shall be applied to extract a binary image, which can be considered as a segmentation method distinguishing the retina from the surrounding dark area. The image shall be cropped using the binary image, which removes the black borders surrounding the image, leaving only the region of interest where the retina can be found. If the image model is in grayscale, no conversion shall be required. Once cropping is complete, the green channel is then extracted from the original image, which is RGB. The selection of the green channel is due to better contrast between retinal vessels and the background with respect to other channels such as red and blue, which makes this channel more effective in enhancing important retinal features. This image, which is the green channel, is then used to perform the next operations involved in the described process.

Algorithm 2 CLAHE and Illumination Normalization

Input:

(1) GreenImage Output:

EnhancedImage

```
1: start algorithm
2: claheImage = applyCLAHE(GreenImage, clipLimit, tileGridSize)
3: blurImage = gaussianBlur(claheImage, sigma)
4: normImage = addWeighted(claheImage, alpha, blurImage, beta, gamma)
5: EnhancedImage = stack(normImage, normImage, normImage)
6: EnhancedImage = resize(EnhancedImage, 224 × 224)
7: end algorithm
```

B.CLAHE AND ILLUMINATION NORMALIZATION

The input to this algorithm is provided in green channel of retinal images, it is enhanced further by using a set of operations. Initially, Contrast Limited Adaptive Histogram Equalization is used to enhance the local contrast. By doing that features such as lesions and blood vessels in the retina get more attention. The amplification of noise is addressed by CLAHE through contrast limiting. After that, a Gaussian smoothing operation is carried out upon the enhanced image to get an estimate for the background illumination. Illumination correction is done by combining the enhanced image obtained using the CLAHE method with the estimate for the background image through a weighted subtraction method. Through the normalization step obtained in the final image, the lighting conditions are rendered more uniform. For appropriate consistency with convolutional neural networks, the normalized image is duplicated across three color channels to obtain an RGB image representation. Finally, the

output image is resized into a fixed dimension of 224x224 pixels for deep learning inputs to generate the final enhancement image.

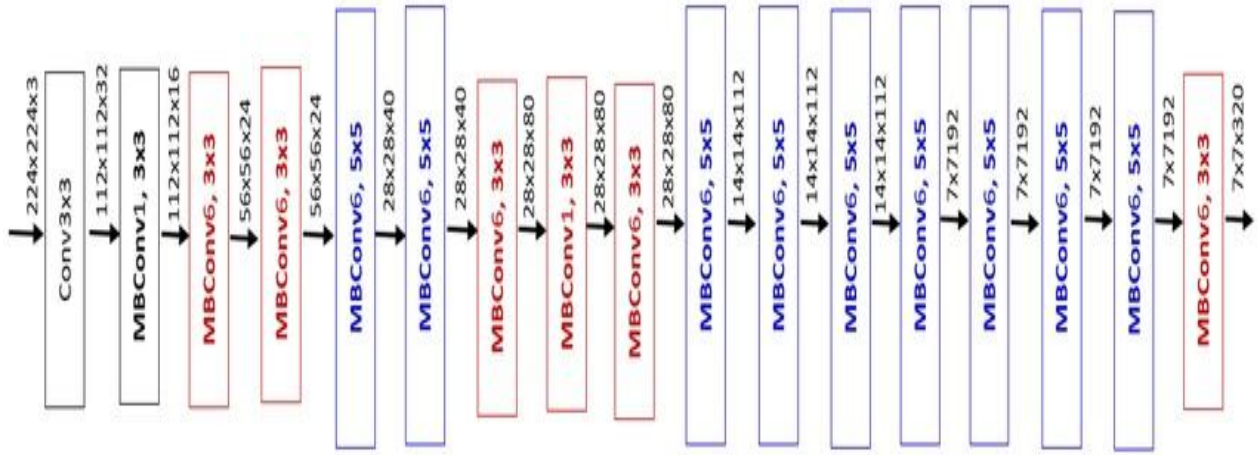


Figure 7: Efficient Net-B2 neural network architecture

III.FEATURE EXTRACTION AND CLASSIFICATION

The proposed structure for the classification of diabetic retinopathy (DR) uses the pre-trained model EfficientNet-B2, combined with data organization, data pre processing, and data imbalance algorithms. The datasets are divided into the training set and the validation set, along with class-wise directory structures to maintain data consistency. A data loader is defined to load the retinal images, convert them to RGB, and assign the correct class labels to the images. For consistent input data, all images are resized into 260x260 pixels, and the standard ImageNet mean and std values are applied for data normalization. Data augmentation strategies such as random vertical and horizontal flips, as well as slight transformations, are only employed during the training phase to promote more diversity as well as prevent overfitting. Class imbalance issues, which are common with medical imaging datasets, are handled by computing the inverse frequency weights of the classes involved. These are subsequently included in the model through a weighted cross-entropy loss function. The EfficientNet-B2 will act as the feature extractor, allowing it to learn low-level texture characteristics as well as high-level characteristics of the retinal images. Also, a new classification head will be implemented, replacing the prior one, to include dropout layers that avoid overfitting, followed by a fully connected layer to suit five-class diabetic retinopathy classification.

IV. DATASET DESCRIPTION

Diabetic retinopathy is a an eye disease cause due to damage in blood vessels in the retina, generally resulting from prolonged elevation of blood glucose levels associated with diabetes. If left unattended, it causes loss of vision and even blindness. DR progresses through different stages, preliminary from gentle abnormalities to severe damage, where new, fragile blood vessels form on the retina, leading to complications. The dataset is divided into folders regarding the presence and/or severity of diabetic retinopathy. The "NO_DR" folder contains images of retinas without any signs of diabetic retinopathy. The main folder "DR" is further divided into four categories based on the severity in condition:

"Mild1" consists of the pictures with an early stage of DR, "Moderate2" involves pictures with moderate destruction to the retina, "Proliferative3" describes the advanced stages where vessels have sprouted abnormally, and "Severe4" consists of pictures taken in a stage showing complete evident damage along with heavy vision loss. These categorized folder structures enable specific treatment analysis along with the possibility of classification into the different stages of the disease effectively. All the images were gathered from APTOS 2019 Blindness Detection dataset, sorted according to the given description.

TABLE 1 The outcomes for Aptos dataset

Method	Accuracy
--------	----------

Classification without enhancement	77%
Classification with enhancement	85.8%

TABLE 2. Comparison with Other models and enhancement using Aptos dataset

S.No	Model Name	Enhancement technique	Accuracy
1.	ResNet-50	Circular crop, Gaussian blur	78%
2.	ResNet-50	Green channel, clahe, Illumination	80.4%
3.	EfficientNet-b2	Circular crop, Gaussian blur	79%
4.	EfficientNet-b2	Greenchannel, clahe, Illumination	85.8%

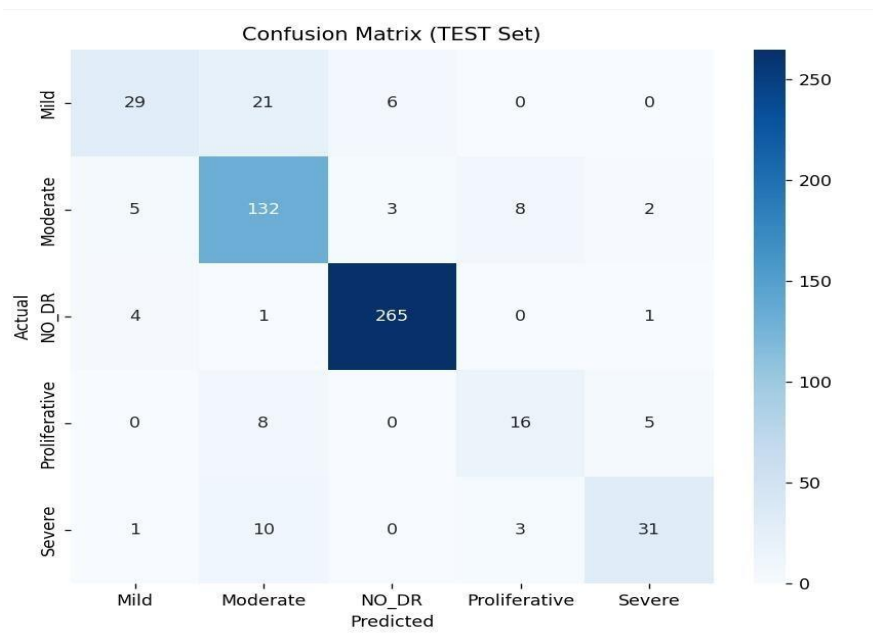


Figure 8: Confusion Matric for the proposed model

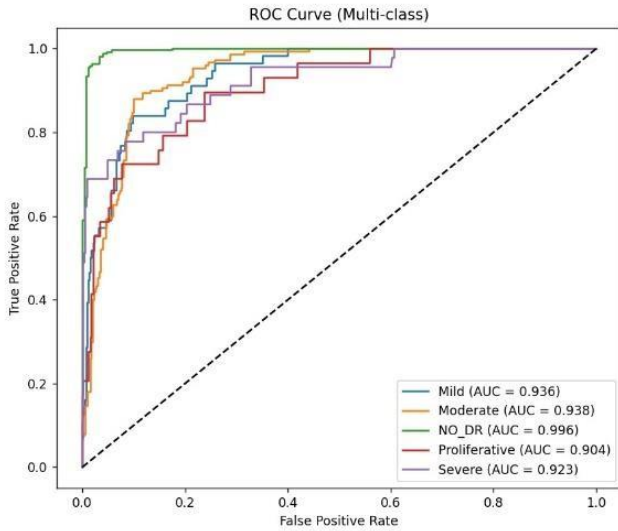


Figure 9: ROCcurve for proposed model

V. CLASSIFICATION REPORT:

The proposed model yields an accuracy of 85.8%, reflecting significantly high general performance. For class-specific evaluation, it yields a precision of 77.3%, indicating that a large proportion of the positive cases were correctly predicted. The recall of 72.3% shows that it could detect a high number of actual positive cases. Besides, the F1-score of 74.5% shows a good balancing in the trade-off between precision and recall; thus, the efficiency and robustness of the model are confirmed.

VI. EVALUATION

The EfficientNet_B2 as pre trained model used to evaluate the test data with images from the APTOS retina images test dataset. The images in test data are placed in respective folders according to classes. Finally, images with dimensions 260x260 pixels and in RGB format with data standardization using ImageNet statistics. The images are read in batches of fixed sizes without any shuffling. The model that is already trained is used to predict values with weights using forward propagation. The overall accuracy and precision, recall, and F1-score values in macro-averaged values are used to evaluate the performance of the model. The overall accuracy of model that is 85.8%. Precision is found to be 77.3%, recall is found to be 72.3%, and F1score is found to be 74.5%. The confusion matrix is provided to visualize the final model performance. Multi-class ROC curves are provided to depict the discrimination ability of the model in the classification of Diabetic Retinopathy (DR). High AUC values, particularly for the No_DR class, suggest that there is good class-wise separation by the proposed model.

VII. DISCUSSION

The results of this study have shown that the image enhancement techniques that were used, such as green channel extraction, CLAHE, and illumination normalization, were effective in enhancing the quality of retinal fundus images, thus making the pathological features in the images prominent. Using these enhanced images for classification using the EfficientNet-B2 deep learning network results in better image classification capabilities. This is because the images are enhanced, thus making the classification of the images more accurate. The results achieved using the classification network suggest an accuracy of 85.8%, thus making the combination of image enhancement techniques and deep learning effective in diabetic retinopathy classification. There is, however, some misunderstanding regarding the distinction between mild and normal cases. This happens because, in the early stages of the illness, the pathological alterations are imperceptible and closely mimic typical retinal patterns, making them hard to tell apart.

VIII. CONCLUSION AND FUTURE WORKS

This work presents a hybrid framework for the detection of diabetic retinopathy that combines illumination normalization, contrast-limited adaptive histogram equalization (CLAHE), and green channel extraction within the EfficientNet-B2 deep learning model. By increasing contrast and reducing illumination variations, the suggested preprocessing technique improves retinal image quality, allowing for more precise feature representation and classification. Performance evaluation on the APTOS dataset showed that the suggested method is reliable and effective, with an overall accuracy of 85.8% and notable gains in

precision, recall, and F1-score. These results demonstrated how crucial it is to improve the deep learning-based diagnostic models for automated retinal screening. To lessen reliance on manual tuning parameters and improve adaptability to various image conditions, future research could expand the framework to incorporate a fully automatic enhancement mechanism utilizing deep learning methodologies. In order to improve early-stage disease detection, provide explainable predictions, handle class imbalance by using GAN-based data augmentation, and increase model robustness, lesion-level segmentation and localization may be combined. Finally, the present system can be further extended to support multi-disease classification in the retina and implemented with real-time mobile or IoMT-based healthcare platforms for large-scale and practical screening applications.